

Tasfi Fairoz Nidhi

✉ tasfi.f.nidhi@gmail.com | 📞 +880 1781-910404 | 🌐 tasfifairoznidhi | 🌐 fairoznidhi | 🌐 Portfolio

RESEARCH INTERESTS

Machine Learning, Network and Systems Security, Trustworthy and Explainable AI, Privacy in Machine Learning, Network Protocols

EDUCATION

Rajshahi University of Engineering and Technology (RUET)

January 2020 – June 2025

Bachelor of Science in Computer Science & Engineering

Rajshahi, Bangladesh

- **CGPA: 3.84/4.00** (3.96 in final 4 semesters) — 8th Position
- **Thesis:** Precision Weed Classification: Leveraging Transfer Learning and Attention Mechanisms for Advanced Weed Detection

PUBLICATIONS

Triple Attention MobileNetV3: Harnessing Integrated Attention and Transfer Learning for Next-Generation Skin Lesion Detection

Conference Paper, 2024

Tasfi Fairoz Nidhi, Anwar Hossain Efat, S. M. Mahedy Hasan, Md. Shahid Uz Zaman, Abdun Nur Wasit

2024 IEEE International Conference on Computing Applications and Systems (COMPAS) – DOI: 10.1109/COMPAS60761.2024.10796988

Dual Concatenated Transfer Learning with Attention Fusion: An Ensemble-Enhanced Approach for Skin Lesion Classification

Journal Paper, 2026

Probal Bhowmick, Julia Rahman, Anwar Hossain Efat, Tasfi Fairoz Nidhi, Dipanjan Karmaker Amit

DIGITAL HEALTH (SAGE) — Q1 Journal – DOI: 10.1177/20552076261458948

WORK EXPERIENCE

Vivasoft Limited

November 2024 – Present

Software Engineer L-1 | July 2025 – Present

Dhaka, Bangladesh

- Building Novara, an AI-powered recruitment platform with an employer side (role-based access control, employee management, job posting, candidate tracking) and a candidate side (AI interview platform, AI resume builder). **Technologies:** React, TypeScript, Tailwind CSS, Redux, RTK Query, TanStack Router.

Associate Software Engineer (Internship, Part-time) | November 2024 – June 2025

Rajshahi, Bangladesh

- Built **VivaMeal**, a daily meal management web app enabling users to update meal preferences and track penalties/food wastage, with admin-side tools for monitoring statistics. **Technologies:** Next.js, TypeScript, TanStack Query, Tailwind CSS.

PROJECTS

HTTP/1.0 Server from Raw TCP Sockets

(GitHub Link)

- Built an HTTP/1.0 server in JavaScript using Node's net module, manually implementing TCP socket handling, raw request parsing, header processing, and a custom router without any HTTP framework. **Technologies used:** JavaScript, Node.js, TCP/IP, HTTP protocol.

Custom 3-bit CPU from Scratch

(GitHub Link)

- Designed and simulated a 3-bit CPU at the gate level (ALU, register file, RAM, program counter, control unit), verified with a working demo running real machine code.

OTHER EXPERIENCES

Competitive Programming

Online

Active Participant

2020 – 2024

- **Codeforces:** _nidhi_ – **Specialist**, Max Rating: 1430, 450+ Problems Solved
- **Other Platforms:** 200+ Problems Solved across CodeChef, AtCoder, LeetCode, LightOJ, and Uva

SKILLS

- **Programming Languages:** Python, Go, C/C++, JavaScript, TypeScript
- **Database Systems:** PostgreSQL, MySQL, MongoDB
- **Networking & Protocols:** HTTP Protocol, Socket Programming, REST APIs, TCP/IP
- **Machine Learning & Deep Learning:** ANN, CNN, Supervised & Unsupervised Learning, Feature Engineering, Transfer Learning, Attention Mechanisms, Computer Vision
- **ML Frameworks & Tools:** TensorFlow, PyTorch, Keras, OpenCV, scikit-learn, NumPy, Pandas
- **Web & Mobile Development:** React, Next.js, React Native, HTML, CSS, Tailwind CSS
- **Other:** Data Structures, Algorithms, Problem Solving, Git

AWARDS

MBSTU CSE IDPC, 2023 – 16th Position

6th National Girls Programming Contest, 2022 – 28th Position

RUET CodeSmash Freshers Programming Contest, 2020 – 10th Position

ACCOMPLISHMENTS

- IELTS Score: 7.0 (L: 7.5, R: 7.0, W: 6.5, S: 6.0).

ACTIVITIES

Secretary, Onuronon – RUET Cultural Club, Rajshahi, Bangladesh

Joint Secretary, Oviatrik – RUET Adventure Club, Rajshahi, Bangladesh